CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 1: Intelligent Agents & State-Space Search A*

Course: Program Elective Course-I [23CS3507]

Category: Assignments | Assignment #1 | Academic Year: 2026-27

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1. Overview & Objective:

Foundations of intelligent agents, heuristic search, and game tree pruning.

2. Assignment Problems & Instructions:

1. Categorize PEAS Performance, Environment, Actuators, Sensors for autonomous vehicle agents.
2. Prove mathematical condition for A* heuristic admissibility and consistency monotonicity .
3. Execute A* algorithm step-by-step on 8-puzzle game calculating Manhattan distance heuristics.
4. Apply Minimax algorithm with Alpha-Beta pruning on a 4-level adversarial game tree.

Submission Guidelines & Verification:

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